Agrivive

Buyer, Seller and System Flows

Every step from registering to posting on social media, ordering, pickup, rating and reporting. Written in plain words, from the code as of 2026-11-08.

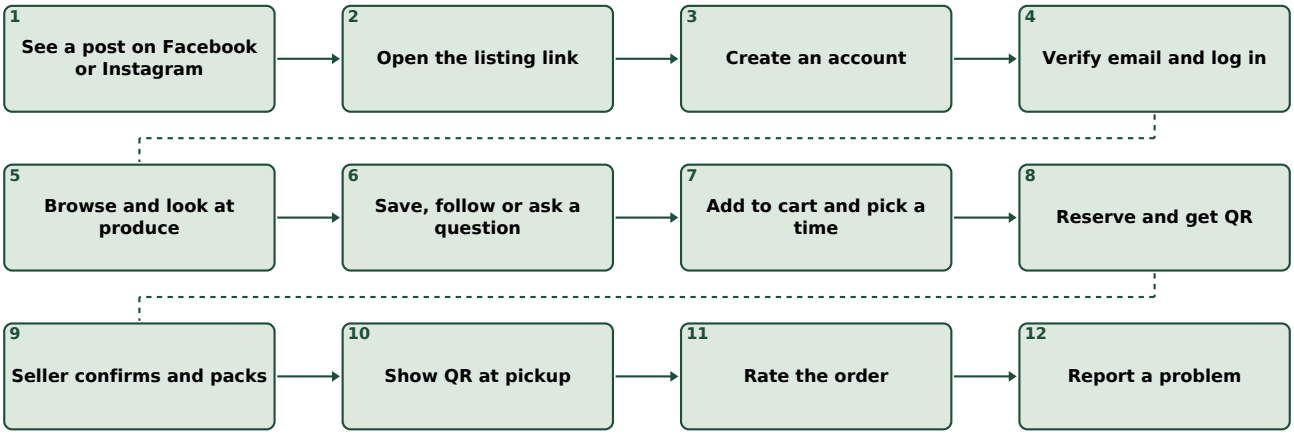
Part 1	Buyer flow: from a social post to the report
Part 2	Seller flow: from registering to posting to scanning
Part 3	System flow: what runs behind the screens

Notes

The buyer reserves produce and pays at pickup. The app does not take payment. Buyers see no rank or score on any screen. Not yet built: the Priority/Standard/Basic tier label (60 and 40 cut-offs) and a direct buyer alert when a price drops. Boost price drops reach buyers through the Facebook and Instagram post. Status lists come from the code, not from a phone test.

Part 1. Buyer Flow

At a glance



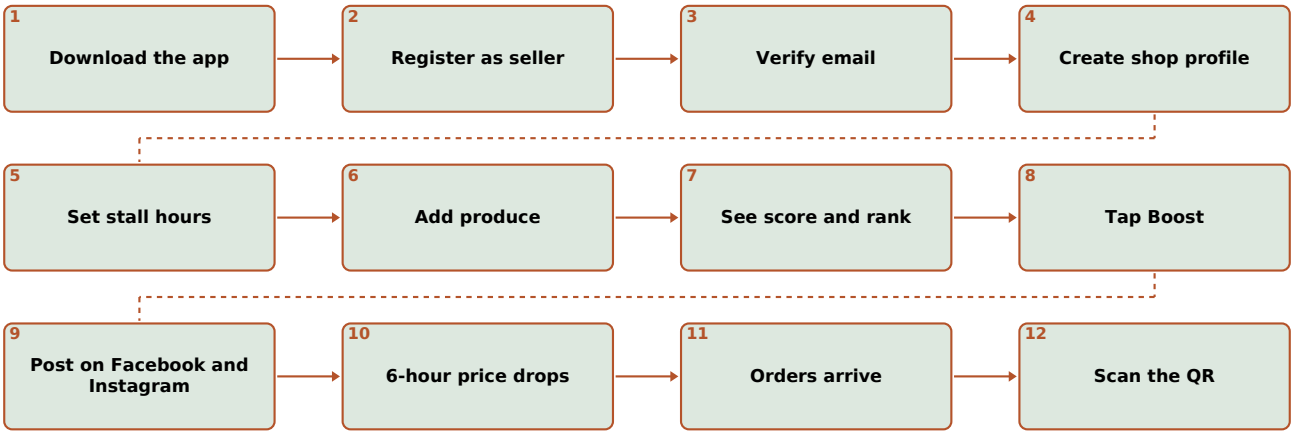
Step by step

#	Step	What the person does or sees	What the system does
1	**See a post on social media**	A buyer sees a Boost post on the Agrivive Facebook Page or Instagram: title, short text, photo and an "Order here" link.	n8n writes the text with Groq. The backend posts to Facebook and Instagram and saves the post record. The link goes to the listing page.
2	**Open the listing link**	The link opens the produce page on the website (/marketplace/[id]). A visitor can look without an account.	Public product read: name, photo, price, amount left, seller, pickup place.
3	**Create an account**	Sign up with an email code or with Google (/signup). Buyers choose the buyer role.	Better Auth sends an email code. /auth/check-email and /auth/register-role set the role. A person can hold both roles.
4	**Verify email and log in**	Enter the code (/verify-email), then log in (/login). Google users can add a password (/password-setup). Lost access goes to /account-recovery.	Session is created. Password status route tells the screen if a password exists. Recovery uses a request and a verify step.
5	**Browse the marketplace**	Open /marketplace for produce, deals, the overview page and the seller map. The buyer sees no rank and no score.	Listings are ordered by the AHP-TOPSIS visibility ranking. A hidden sort value is sent but never shown.
6	**Look at produce and seller**	Open the produce page and the seller page (/sellers/[id]): photos, price drop, stock, rating, reviews, map.	Reads product, seller, rating and review routes.
7	**Save, follow, ask**	Tap the heart to save (/favorites). Follow a seller to get new-listing notices. Ask a question about the produce, or open Messages.	Wishlist and follow routes. Product inquiries and chat conversations (text and attachments). Followers get a notice when a seller posts new produce.
8	**Add to cart and choose a time**	Put produce from one or more stalls in the cart (/cart). Pick a pickup time range for each stall (Today or Tomorrow).	Checkout (POST /checkouts) splits the cart by seller. Pickup slots come from each stall's hours, capacity and pickup range. Closed or away stalls cannot take orders.
9	**Reserve and receive a QR**	The order is a reservation, not a payment. The buyer gets an order page with a QR code and a status.	Order is created as pending. Stock is taken away for the reservation. QR is valid to the end of the pickup slot plus 15 minutes and cannot be extended. A new order gives a new QR.

#	Step	What the person does or sees	What the system does
10	**Seller confirms and packs**	The buyer sees the status move: confirmed, packed, ready. A notice arrives in the app and by push.	Seller confirms, packs and marks ready. If the seller does not confirm, the worker auto-confirms after 3 hours. A notice goes out about 1 hour before the reservation ends.
11	**Change or cancel**	The buyer can cancel (POST /orders/:id/cancel), answer a reschedule request, or send an excuse or decline note.	Cancelled stock is put back. Cancel, decline, excuse and reschedule are recorded for trust points.
12	**Show the QR at pickup**	The buyer shows the QR to the seller at the stall in the chosen time range.	Seller scans it. Order becomes completed. Scanning only takes quantity away, never adds it back.
13	**Not picked up**	If the buyer does not come, the order ends.	Order becomes expired or a no-show is recorded. A buyer no-show adds 2 trust points. Points count for 90 days.
14	**Rate the order**	After pickup the buyer rates once per order: produce stars, seller stars, an optional comment and one optional photo. It can be edited for 15 days.	Seller rating = average of seller stars. Produce stars never change trust or ranking. Rating feeds the seller reliability criterion.
15	**Report a problem**	Open the order and tap report. Reasons: pickup problem, conduct, listing not accurate, QR misuse, other. Evidence can be added.	A report needs 2 different people to confirm. The system decides the case by its rules, adds trust points and can pause or restrict an account.
16	**Everyday account tasks**	Profile and photo (/profile), notices (/notifications), messages (/messages), app download page.	Notices can be marked read. Inactive users are checked by the worker.

Part 2. Seller Flow

At a glance



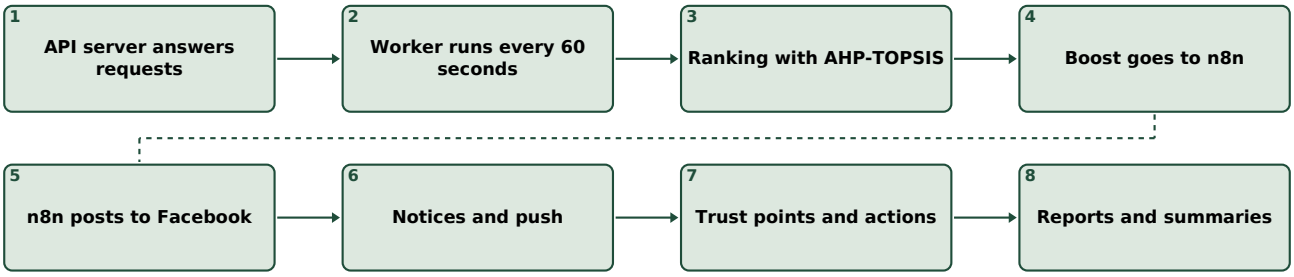
Step by step

#	Step	What the person does or sees	What the system does
1	**Get the app**	A seller installs the Agrivive seller app (Expo, Android) from the download page.	The mobile app talks to the same API as the website.
2	**Register as a seller**	Sign up with an email code (or Google) and choose the seller role.	/auth/check-email and /auth/register-role. The same email can also be a buyer.
3	**Verify email**	Enter the code and log in.	Session starts. Many seller actions need a verified seller.
4	**Create the shop profile**	Fill in shop name, address, phone, and the shop place on the map. Add a photo.	GET /seller/setup shows what is left. POST /seller/profile saves the profile. The place is used by the seller map.
5	**Set up the stall**	Set opening hours for each day, pickup range, how many pickups at once, Closed today, and Away mode.	Rules feed the pickup slots buyers see. A job ends Away mode when its time is over.
6	**Add produce**	Pick the vegetable from the catalog, add a photo, price, starting price, amount and unit (sack, kilo or pile). Optional: price cut in pesos and the lowest price.	POST /seller/products. A listing cycle begins and publication time is saved. Followers get a new-listing notice. Duplicate listings are guarded.
7	**Check visibility**	The Visibility tab shows a score out of 100, a rank, and the four things that count: time posted, price drop, amount left, reliability.	AHP-TOPSIS ranks all eligible listings. Weights are 0.3507, 0.3507, 0.1892 and 0.1093.
8	**Tap Boost**	After adding produce the app asks Boost or Not now. Boost posts the listing on the Agrivive Facebook Page and Instagram. Only one hand-made Boost per listing.	POST /seller/products/:id/boost queues a boost. The worker sends n8n only the boost id and the reason. A paused seller or a high fault rate cannot Boost.
9	**Posting happens**	The seller sees the status: queued, posted, or try again.	n8n asks the API for the current facts, Groq writes the title and text, the backend posts to Facebook and Instagram and saves a pending row first so a crash never posts twice.
10	**6-hour price drops**	6 hours after the post the seller sets the peso drop and final price (if not set). The price then falls every 6 hours until it reaches the final price.	Worker sends the follow-up to n8n. The API lowers the price, n8n posts a flash-sale caption. Stops if sold out, archived, paused, or at the floor. Seller gets a notice each time.

#	Step	What the person does or sees	What the system does
11	**Auto-post (optional)**	The seller can switch on posting again when the produce ranks high.	Works only when the server flag AUTO_BOOST_ENABLED is on. One ranked boost per round. Listings boosted in the last 6 hours are skipped.
12	**Orders arrive**	A notice says a new reservation came in. The seller opens Orders, confirms, packs and marks ready.	Stock was already reserved. Declines, excuses and reschedule requests can be answered. Buyer questions must be answered: reminder at 12 hours, warning at 24, flagged at 48.
13	**Scan the QR at pickup**	The buyer shows the QR. The seller scans it (or opens the order) and completes the pickup.	POST /seller/orders/scan checks the QR and the time. The order becomes completed. Seller can also mark a no-show.
14	**Manage stock**	Add More, Take Away, Change Amount, and see Change History. Old stock over 6 hours old triggers a new-harvest warning.	Adjustments are saved with a history. Stock at zero removes the listing from buyers.
15	**Read the sales report**	Open Analytics: Sold, How Much Sold (sell-through), Where Your Produce Went, Pickup Results, Stock Right Now, and a short summary.	Sell-through = picked up ÷ posted × 100. A Groq summary is made from the numbers.
16	**Reviews, trust and alerts**	See ratings and reviews, the trust standing page, and notice settings (Stock and Price, Boost, Account, Night Mode).	Points: no reply 1, seller cancels 2, low rating 1, misleading info 2, rude message 2, QR misuse 3. Warning at 4, restriction at 6, 8, 10, block at 12.
17	**End of a batch**	Archive the listing, or reactivate and post again.	Archive hides it from buyers. A new batch starts a new cycle, and the old 6-hour rule ends.

Part 3. System Flow

At a glance



Step by step

#	Step	What people notice	What the system does
1	**Two programs, one codebase**	Not visible to users.	The API (src/index.ts) answers the apps and the website. The worker (src/worker.ts) runs the scheduled jobs every 60 seconds. Run exactly one worker. The API also checks push alerts and listing notices every 30 seconds, so those jobs must be safe to run twice.
2	**Login and roles**	People sign in with an email code or Google.	Better Auth with an Expo plugin for the phone. Roles are admin, buyer, seller and stakeholder. Protected routes check the role and the account status, and apply trust blocks.
3	**Marketplace ranking**	Buyers see a list in a fair order.	AHP-TOPSIS: four criteria (posting age, price drop, stock left, seller rating), AHP weights from a Saaty matrix with consistency ratio 0.0038, then TOPSIS closeness score. Scores are relative to the other listings.
4	**Reservations**	A buyer reserves. A seller confirms.	Order create reserves stock inside a database transaction. The QR carries the order and slot. Seller scan completes the order. Stock changes never add back on scan.
5	**Worker: order jobs**	Orders move on without a tap.	Auto-confirm after 3 hours, expire pending orders, record unmarked no-shows, auto-accept declines, end Away mode, notice about 1 hour before a reservation ends.
6	**Worker: trust jobs**	Warnings and limits arrive on time.	Inquiry deadlines (12h reminder, 24h warning, 48h non-responsive), send trust notices, end restrictions when their time is over, recalculate standings, check inactivity (3 weeks warning, 1 month block).
7	**Worker: listing and boost jobs**	Boosts and price drops run on schedule.	Send new-listing notices to followers, queue a ranked boost (off unless enabled), deliver queued boosts to n8n, deliver 6-hour follow-ups (retry every 15 minutes, up to 5 times), send price-drop setup notices.

#	Step	What people notice	What the system does
8	**n8n workflow (12 nodes)**	A post appears on Facebook and Instagram.	Boost Confirmed webhook, Follow-Up Call?, Get Boost Context, Already Posted?, Write Caption (Groq), Format Caption, Post to Facebook, then the follow-up path: Get Latest Context, Reduce Price, Price Reduced?, Flash-Sale Caption, Format, Post. n8n receives only ids. It calls back with a Bearer service token.
9	**Rules the backend owns**	Posts are not doubled and not too frequent.	6-hour lock after a boost, 3 hours between price-drop posts, a pending row saved before each post, price never below the seller's final price. The 6 hours are counted by the backend, not by n8n.
10	**Notifications**	People get alerts.	App notices, push (FCM) and email (Resend or Mailtrap). Alerts of one kind in one 60-second window become one alert. Sellers can turn off groups and use Night Mode.
11	**Trust and reports**	Fair treatment for buyers and sellers.	Points last 90 days. A report needs 2 different people to confirm. Points and actions are gated by TRUST_ACTIONS_ENABLED. Recovery and dispute routes exist.
12	**Analytics and summaries**	Sellers, the admin and stakeholders see numbers.	Seller metrics (sell-through, sold, expired, cancelled), a Groq summary, an admin performance page and a stakeholder summary.
13	**Files and secrets**	Photos load safely.	Photos live in S3. Secrets (database, Meta, FCM, S3, mail, n8n tokens) stay in the server .env. Web uses only public settings.
14	**Running it**	The site stays up.	Docker Compose on a VPS: Caddy, API, worker and web. Database changes are applied by hand as a release step and never on startup. Plugins add security headers, rate limiting and request validation.